Clinical Imaging & Verification Report

Uncertainty-Calibrated & Adaptive Self-Verifying Pulmonary Assessment

REPORT
2026-10-19

PATIENT & SCAN PROFILE

PATIENT ID	SCAN DATE	SCAN FORMAT	SLICES
Lung_Dx-B0041	2026-10-19	Real DICOM Slice (C:\Users\spide\Desktop\LUCID-CT\1.dcm)	16

ESTIMATED CATEGORY

4A Suspicious

Calibrated Malignancy Risk: 47.2%

RISK TIER

INTERMEDIATE

Estimated diameter: 10.0 mm

ADAPTIVE VERIFICATION & RELIABILITY GATE

ADAPTIVELY_RESOLVED (CRI: 78.8%)

ADAPTIVELY_RESOLVED

Recommended Clinical Action

VERIFIED via Adaptive Re-Analysis: Initial inconsistency investigated and resolved with high multi-scale agreement (95.0%) and CRI (78.8%). Proceed with guideline-based clinical management.

TARGETED RE-ANALYSIS AUDIT TRAIL

Agreement with Baseline: 95.0%

* 1.25x High-Magnification Boundary Shell Zoom

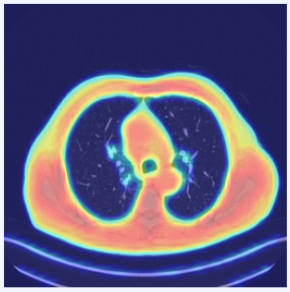
UNCERTAINTY & VERIFICATION INDICES

Epistemic Uncertainty	**0.0108**
Aleatoric Uncertainty	**0.6706**
Predictive Entropy	**0.6927**
Scan Quality Score	**73.9 / 100**
Grad-CAM Fidelity	**0.66 / 1.0**
Trait Physical Grounding	**0.69 / 1.0**

RADIOLOGICAL TRAIT EVIDENCE

Spiculation	**2.31**
Lobulation	**2.16**
Calcification	**3.46**
Subtlety	**3.38**
Margin	**1.93**
Sphericity	**2.28**

XAI LOCALIZATION & EVIDENCE GROUNDING



Estimated from diameter alone, assuming a baseline solid nodule. Nodule type, growth, and prior imaging were not available.